ADAGIO GROUP

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Reply to responses on paper 1

Benjamin D. Summers <bsummers@adagiollc.com> Sat, Sep 26, 2026 at 12:49 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Ezequiel,

Thank you for the second round. The revised deposit is at https://dx.doi.org/10.2139/ssrn.6922380 and carries every amendment in items 1, 3, and 4. Nothing in item 2 changed. Below are the responses in your order.

1. The report. The corrected report is welcome, and the two prescriptions you withdraw are corrected in the text. Section 4 ends the time-slice sentence at "the axioms assign no algebra to the sharp surface itself" and draws nothing from it; the next sentence says the framework assigns content to thin regions and has no object at a sharp time, which is the indexing fact and no more. Definition 1(a) now says "uniquely solvable," and the abstract says the same in place of "well-posed." I reproduced your counterexample before writing this: the stationary law is (0.264, 0.348, 0.388), the rates into the first state from the merged pair are 0.5 and 0.3, and the lumped family's defect at (0, 0.9, 1.8) has spectral norm 0.0024979. It sits in the second branch of the lemma as now stated. It is not in the paper; the paper reports no computed quantity and I am keeping it that way. The sampling floor is in Section 6: the lemma bounds from above and gives no lower bound, so an application must declare the window width below which the estimated defect is not distinguishable from sampling error at the sample size in hand, calibrated by simulation or bootstrap rather than derived, and the test is silent below it. Your conditional heuristic for $\delta_{\min}$ is not in the paper.

3. The lemma. Accepted on all three points. The statement now reads: the kernels are the regular conditional distributions of $Y_\delta(u)$ given $Y_\delta(s)$, and the total variation of $\Delta_\delta(\cdot|y)$ is at most $C\, q_* \,\delta$ for almost every y with respect to the law of $Y_\delta(t_1)$. The proof names μ_y and $\nu_{y'}$ as regular conditional distributions, defined almost everywhere with respect to the laws of $Y_\delta(t_1)$ and $Y_\delta(t)$, and states that the bound $\max_x \Sigma_z |D_x(z)|$ holds for almost every y because it is uniform in x and μ_y is a probability measure on S. The constant is $2 + 8/c$ as before. The rate is "at least linearly" in the abstract, Section 2, and Section 6; the standard from Theorem 5.1 of the convex-perturbation paper applies here and I had not applied it. Section 2 now makes persistence conditional: when the limiting lumped defect is nonzero, no refinement removes it. The lemma's "generically positive" is unchanged, and the sentence after it in Section 6 already says that a vanishing defect does not establish the Markov property, by Feller.

4. The four amendments. Accepted. Section 2 says the state at t is the limit of interval averages under a stated observation model and names the uniform average of equation (1) as the model used. In Section 4 the regulator is a momentum cutoff Λ_0, the Wilsonian paragraph runs on Λ_0 throughout with $\Lambda_0 \to \infty$ for the continuum limit, Λ is the validity scale only, asymptotic freedom is one route to an interacting continuum limit with an interacting ultraviolet fixed point as the other, the bare theory at a point is described as the $\Lambda_0 \to \infty$ end of the flow rather than as the boundary of validity, and the abstract no longer identifies the divergences with finite validity. On Wightman I went to the source, Ann. Inst. H. Poincaré A 1, 403 (1964), Théorème 3.1 on page 415. It is stated for a field of bounded operators: a function B from spacetime to bounded operators with $B(x)$ in the algebra of every region containing x, translation covariant, commuting at spacelike separation, in a theory with the spectrum condition and a unique vacuum, and its conclusion is $B(x) = b \cdot 1$. The paper now states that version. The unbounded version with the vacuum in the domain was not his theorem and is gone. The minimum-time language is now an obstruction within the semiclassical domain of the bounds in every place it occurred, nine sentences across the abstract and Sections 1, 3, 5, 6, and 7; "cannot be physically implemented," "cannot be reached," and "no measurement can underwrite" no longer appear.

5. Originality. Accepted, both the grading on the foundational axis and the placement of the paper as the premise of the set. On the invariant: I agree it is missing, and I think it is missing for a reason that changes what the problem is. By the fact you cite, augmenting the state with its full history renders any process Markov, so an invariant that is independent of every description cannot exist; what can exist is an invariant relative to a physically fixed class of admissible descriptions, and the open problem is that class. This paper fixes it only in part, by the resolution bound of Section 3 and the non-invasiveness condition of Section 6. Section 7 now says so in one sentence, so the open problem is recorded in the form in which it can be worked on.

Bibliography: since my last email, one entry, Albert 2000, is added to the positioning paragraph of Section 7; this revision adds and removes nothing.

Let me know your thoughts.

Ben

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